

MODEL QUESTION PAPER 1
FIFTH SEMESTER- INSTRUMENTATION ENGINEERING
COURSE- LASER AND OPTICAL INSTRUMENTATION

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	Cognitive Level
1	The bending of light at corners of an obstacle is	M1.01	R
2	In Young's double slit experiment, the wavelength of incident beam should be.....	M1.02	U
3	The process of population inversion is to increase the number of atoms instate.	M1.04	R
4	The type of Laser used in CD and DVD players is	M2.01	R
5	Holography is based on the principle of.....	M2.02	R
6	The principle used in optical fiber communication is.....	M3.01	R
7	In photodiode, when there is no incident light, the negligible reverse current is called.....	M3.04	U
8	 optical fiber sensor modulates light from surroundings.	M4.02	R
9	Write any two advantages of fiber optic sensor.	M4.04	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	State the conditions for bright and dark fringes in interference.	M1.01	U
2.	State the laws of polarization.	M1.01	R
3.	Define metastable state.	M1.04	R
4.	List the medical applications of laser.	M2.03	R
5.	Illustrate the principle of velocity measurement using laser.	M2.04	U
6.	Outline the construction of optical fiber.	M3.01	U
7.	List the advantages of fiber optic communication.	M3.03	R
8.	Explain the construction of photo transistor.	M3.04	U

9.	Describe the general structure of fiber optic sensor.	M4.01	U
10.	Illustrate the working of fiber optic strain sensor.	M4.03	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	Cognitive Level
III	Explain Young's double slit experiment for interference.	M1.02	U
	OR		
IV	Describe the characteristics of Laser.	M1.04	U
V	Describe the experimental set up used to form Newton's rings.	M1.03	U
	OR		
VI	Describe the polarization phenomenon.	M1.01	U
VII	Explain the working of Nd-YAG Laser.	M2.01	U
	OR		
VIII	Illustrate the basic principle of holography.	M2.03	U
IX	Compare single mode, multi – mode and graded index optical fiber.	M3.02	U
	OR		
X	Describe the block diagram of fiber optic communication.	M3.03	U
XI	Describe the operation of Avalanche photodiode.	M3.04	U
	OR		
XII	Explain the working of LED drive circuit.	M3.04	U
XIII	Explain the method of pressure measurement using optical fiber.	M4.03	U
	OR		
XIV	Illustrate the working of fiber optic humidity sensor.	M4.03	U

MODEL QUESTION PAPER 2
FIFTH SEMESTER- INSTRUMENTATION ENGINEERING
COURSE- LASER AND OPTICAL INSTRUMENTATION

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	Cognitive Level
1	The refractive index of a medium is defined as the ratio of speed of light in and the speed of light in the	M1.01	R
2	In interference intensity varies as the	M1.02	U
3	The light wave in which the electric field vector vibrates only in one particular transverse plane is known as	M1.04	R
4	Wavelength of the emitted Laser in a Ruby Laser is	M2.01	R
5	Laser velocity meter works on the principle of	M2.03	R
6	is the thin layer that surrounds the fiber core	M3.01	R
7	Angle of incidence for which the angle of refraction is 90° is known as	M3.01	R
8	Write any two advantages of fiber optic sensors..	M4.04	R
9	fiber optic sensor modulates the properties of fiber.	M4.02	U

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	State the conditions for bright and dark fringes in interfere	M1.01	R
2.	Define dispersion.	M1.01	R
3.	Define stimulated emission.	M1.04	R
4.	List any six industrial applications of laser.	M2.02	U
5.	Illustrate the principle of laser printer.	M2.02	U
6.	Outline the construction of single mode optical fiber.	M3.02	U
7.	Draw the block diagram of fiber optic communication system and write the function of repeater.	M3.03	U

8.	Explain the construction of photo diode.	M3.04	U
9.	Describe the general structure of fiber optic sensor.	M4.01	U
10.	Illustrate the working of fiber optic level sensor.	M4.03	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	Cognitive Level
III	Explain interference phenomenon and state the conditions or bright and dark fringes.	M1.02	U
	OR		
IV	Describe the basic requirements for producing Laser.	M1.04	U
V	Compare interference and diffraction phenomena.	M1.01	U
	OR		
VI	Describe optical pumping and population inversion.	M1.04	U
VII	Explain the working of Ruby Laser with energy level diagram	M2.01	U
	OR		
VIII	Illustrate the method of reconstruction in holography.	M2.02	U
IX	Compare the propagation of light in multi – mode (step index and graded index) optical fiber.	M3.02	U
	OR		
X	Describe the advantages of fiber optic communication.	M3.03	U
XI	Describe the operation of PIN diode.	M3.04	U
	OR		
XII	With diagram explain the working of LED.	M3.04	U
XIII	Explain the method of flow measurement using optical fiber.	M4.03	A
	OR		
XIV	Illustrate the working of fiber optic torque sensor.	M4.03	A